Def. Let $I \subseteq \mathbb{R}$ be an interval, and let $f: I \to [0, \infty)$ be a non-negative valued. We said that $\int_I f$ converges if f is Riemann-Integrable on any $[c,d] \subseteq I$ with

$$\sup\left\{ \int_c^d f \;\middle|\; [c,d] \subseteq I \right\} < \infty.$$

Prop. If a function $f: [0,\infty) \to [0,\infty)$ is Riemann-Integrable on $[0,A]$ for all $A > 0$, then $\int_{[0,\infty)} f = \lim_{A\to\infty} \int_0^A f$.

Proof. Suppose that the improper integral converges.

First, given $A > 0$, $\int_0^A f \le \int_{[0,\infty)} f$ by definition. Since f has only non-negative, so $\int_0^A f$ increases monotonically, so the limit $\lim_{A\to\infty} \int_0^A f$ exists, with $\le \int_{[0,\infty)} f$. //

secondly, by the definition and property of supremum, given $\varepsilon > 0$, there is $[c,d] \subseteq [0,\infty)$ such that $\int_{[0,\infty)} f - \varepsilon < \int_c^d f \le \int_{[0,\infty)} f$.

So, for $A > d$, $\int_{[0,\infty)} f < \int_c^d f + \varepsilon \le \int_0^A f + \varepsilon \le \lim_{A\to\infty} \int_0^A f + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, we obtain

$$\int_{[0,\infty)} f \le \lim_{A\to\infty} \int_0^A f.$$

Consequently, $\int_{[0,\infty)} f = \lim_{A\to\infty} \int_0^A f$. ☐